WHY WILDFIRE?

Next-day wildfire spread prediction from satellite data · Phase 1 · TensorTitans, Section C

Day t+1

Real-world impact

Evacuations and fire crews are planned a day ahead. Responders need tomorrow's fire map, not today's.

13 layers

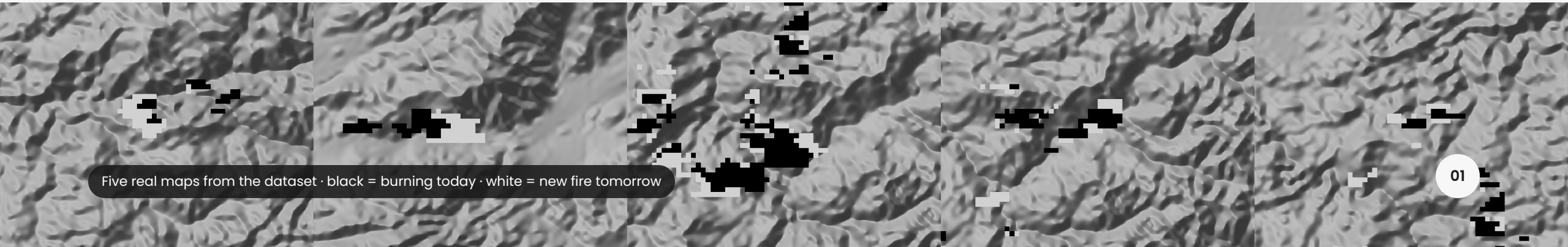
Real, public data

Weather, terrain, fuel and fire for every 1 km square. Huot et al. (2022), IEEE TGRS, US fires 2012–2020.

0.08–1.09%

A hard ML problem

Share of pixels on fire the next day. So rare that accuracy is misleading and neighbours look alike.



Five real maps from the dataset · black = burning today · white = new fire tomorrow

THE DATA

The dataset (Huot et al., 2022) ships as 19 TFRecord files. Each file holds many 64 × 64 km maps, and every 1 km square becomes one table row.

19 FILES



- **15** train · 3,047 MB
- **2** validation · 382 MB
- **2** test · 344 MB

Whole files go to one split, so a fire never appears in two splits.

HOW WE GOT THE ROWS

STEP	TRAIN	VALIDATION	TEST
1 Maps read from the files	30	10	10
2 Squares (maps × 4,096)	122,880	40,960	40,960
3 Remove unobserved squares (-1)	-5,729	-11,168	-8,093
4 Maps with fire: all fire + 10 no-fire per fire	682 10 maps · 62 + 620	429 1 map · 39 + 390	1,747 5 maps · 157 + 1,590
5 Maps without fire: every square kept	76,562 20 maps	25,696 8 maps	12,682 5 maps
Rows used (step 4 + step 5)	**77,244**	**26,125**	**14,429**
of which on fire next day	62	39	157

Sampling only thins maps that contain fire, so fire-free maps supply most rows. One validation map was fully unobserved (9 maps remain), and all 39 validation fires come from a single map.

THE COLUMNS

What is r (train / test)?

The correlation between a column and next-day fire, from -1 to +1, where 0 means no linear link. We compute it twice: on the train rows and on the test rows. A column only counts as useful if r is clearly above 0 in both, like **PrevFireMask .12 / .31**.

How are the groups made?

By type of information, added one layer at a time, so the ablation shows what each type adds:

- **A** Weather on day t — 6 features
- **B** A + fire today — 7 features
- **C** B + land, fuel, people — 12 features
- **D** C + engineered features — 18 features

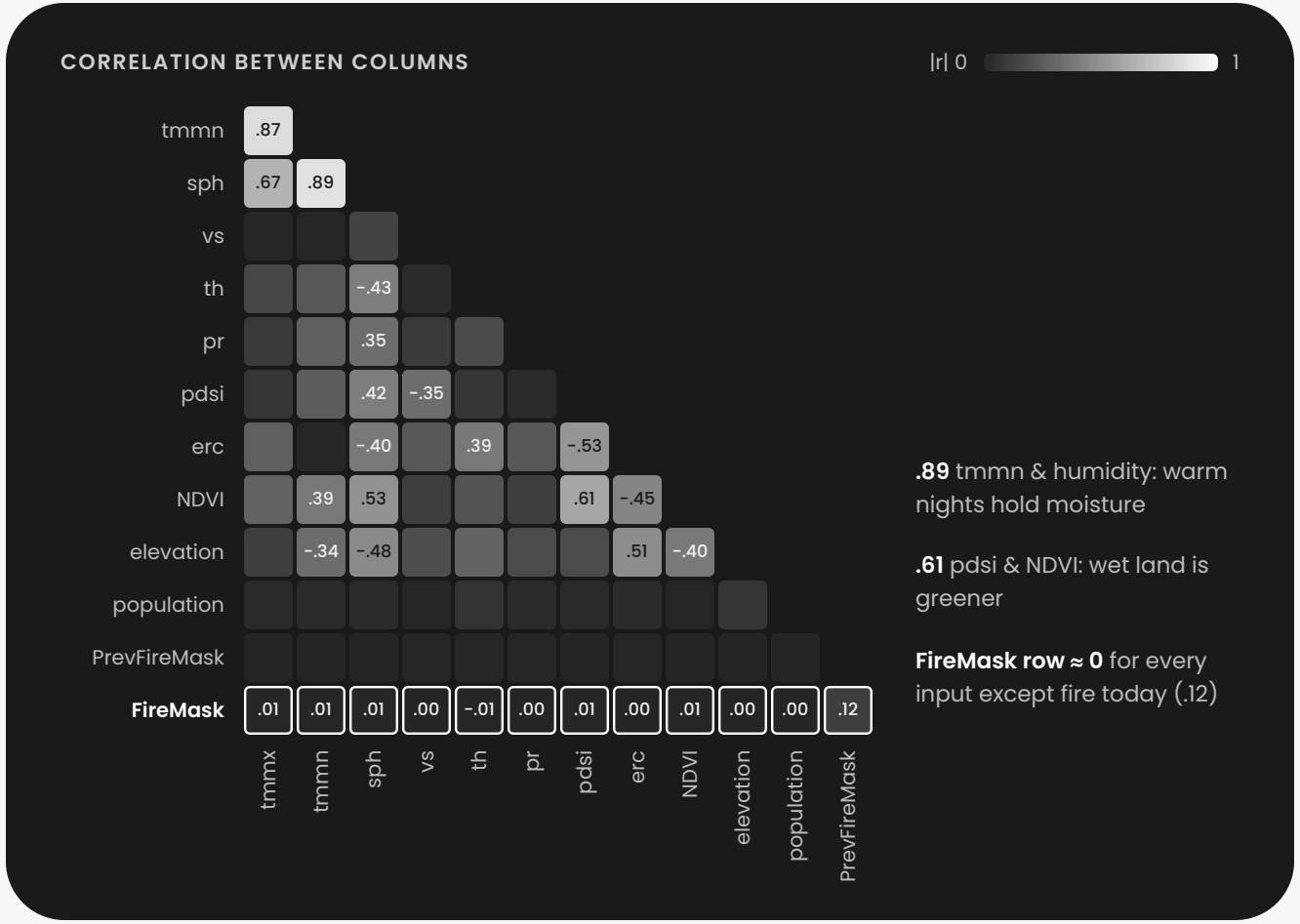
COLUMN	FULL NAME	UNIT	USED FOR	GROUP	R TRAIN / TEST
elevation	Elevation	m	Terrain height; fire runs uphill	C	.00 / .28
th	Wind direction	deg	Which way fire is pushed	A	-.01 / .10
vs	Wind speed	m/s	How hard fire is pushed	A	.00 / -.10
tmmn	Minimum temperature	K	How warm the night stays	A	.01 / -.02
tmmx	Maximum temperature	K	Daytime heat that dries fuel	A	.01 / .07
sph	Specific humidity	kg/kg	Moisture in the air	A	.01 / -.07
pr	Precipitation	mm/day	Rain dampens fuel	A	.00 / -.07
pdsi	Palmer Drought Severity Index	index	Long-term dryness	C	.01 / .07
NDVI	Normalized Difference Vegetation Index	index	Amount of green fuel	C	.01 / -.01
population	Population density	per km²	Human ignitions	C	.00 / -.01
erc	Energy Release Component	index	Heat dry fuel would release	C	.00 / .20
PrevFireMask	**Fire mask, today (day t)**	**0 / 1**	**Is this square burning now**	**B**	**.12 / .31**
FireMask	Fire mask, tomorrow (day t+1)	0 / 1	The label we predict	—	—

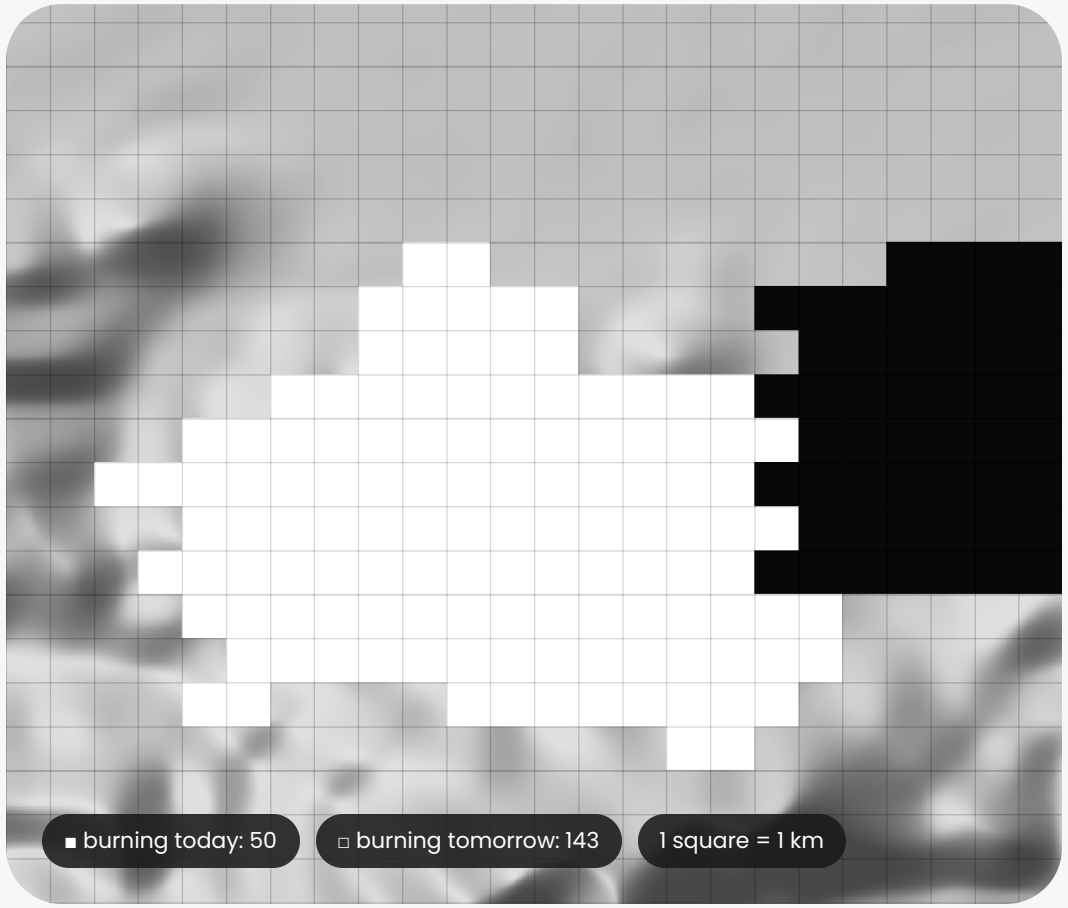
EDA

Exploratory data analysis, train set

AVERAGE VALUE	NO FIRE NEXT DAY	FIRE NEXT DAY
Max temperature (K)	296.7	299.3
Wind speed (m/s)	3.64	3.67
Humidity (kg/kg)	0.007	0.008
Fuel dryness, ERC	38.0	36.0
Burning today	**0.13%**	**16.1%**

Squares that burn tomorrow are **120×** more likely to be burning today, while every weather gap is small. 0 missing values. Based on 62 fire squares.





FEATURE ENGINEERING

Real test map #5 (left): new fire appears right next to existing fire. We turned that into features.

temp_range	tmmx - tmmn: big swing = dry air
wind_u, wind_v	wind as x/y, so 359° and 1° are both north
drought_fuel_index	ERC × drought, combined
fire_neighbor_3x3 / 5x5	**burning squares around each square today**

BURNING NEIGHBOURS, AVERAGE



No fire next day (top) vs fire next day (bottom): **90×** higher. **Bug fixed:** counts were taken after rows were sampled away; now from the full map. r with label: train .065 → .154, test .374.

01 · FLOOR

Persistence

Tomorrow = today. Does learning beat a rule?

PR-AUC	ROC-AUC	
0.0015	0.498	coin flip

02 · LINEAR

Logistic Regr.

Explainable weights. Is a straight line enough?

PR-AUC	ROC-AUC	
0.0008	0.141	backwards

03 · MAIN MODEL

LightGBM

Boosted trees. Do feature combinations help?

PR-AUC	ROC-AUC	
0.0016	0.518	best → test

04 · CROSS-CHECK

XGBoost

Second tree library. Or one library's quirk?

PR-AUC	ROC-AUC	
0.0009	0.199	backwards

THE MODELS

Validation set, 39 fire squares. Random guessing scores PR-AUC 0.0015 and ROC-AUC 0.500. Each model answers one question, from a no-learning rule to boosted trees.

Why LightGBM won

The only model above 0.5 ROC-AUC. Trees combine clues (hot *and* a neighbour burning); **scale_pos_weight = 8** weights the rare fires.

But only just

0.0016 vs 0.0015 for random guessing, so it was chosen for the test, not proven.

FROZEN TEST SET · LIGHTGBM, GROUP D

	ACTUAL FIRE	ACTUAL NO FIRE
PREDICTED FIRE	0 fires caught	4 false alarms
PREDICTED NO FIRE	157 fires missed	14,268 correct

MODEL ACCURACY
98.88%

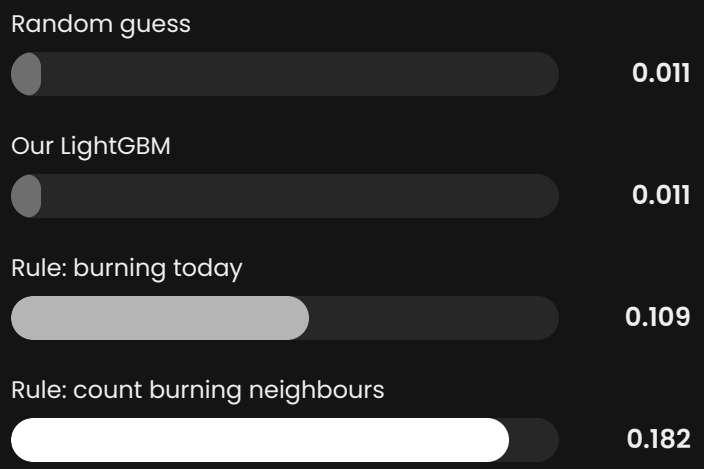
ALWAYS "NO FIRE"
98.91%

THE RESULTS

TEST METRIC	LIGHTGBM	RANDOM GUESS
PR-AUC	0.0109	0.0109
ROC-AUC	0.500	0.500
Recall at 5% false alarms	0.00	0.05

Why not accuracy: 99% of squares are "no fire", so never predicting fire scores 98.91%, higher than our model. PR-AUC and ROC-AUC can't be fooled that way. Threshold tuned on validation only; test used once.

TEST PR-AUC · HIGHER IS BETTER



A one-line rule using our neighbour feature is **17×** better than random (ROC-AUC 0.632 vs the model's 0.500).

KEY INSIGHTS

The signal is in the data

Counting burning neighbours alone reaches PR-AUC 0.182 on test. Our model didn't learn it yet.

Too few fires to learn from

Only 62 fire rows in train. The final model collapsed to 3 distinct output values.

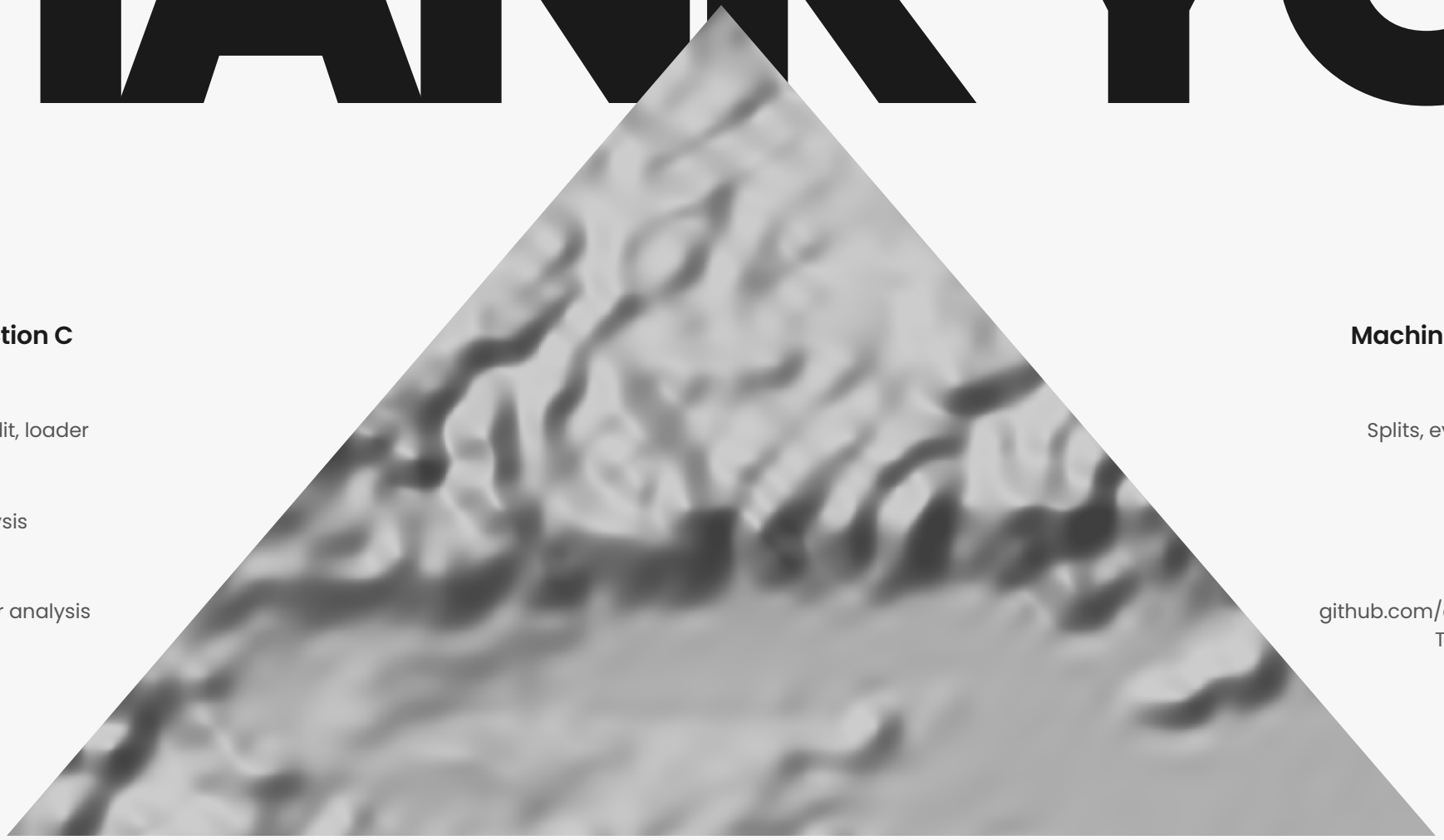
Fire today is the real driver

All 11 weather and land columns have $|r| \leq 0.015$ with next-day fire on train.

Next steps

Hundreds of maps per split, early stopping, threshold tuned on the same model, then Phase 2: a U-Net on full maps.

THANK YOU



Presented by

TensorTitans · Section C

Vanshika
Literature, dataset audit, loader

Riya / Tejas
Exploratory data analysis

Krish
Features, models, error analysis

Phase 1

Machine Learning Project

Nitanshu
Splits, evaluation, comparison

Riya
Phase 1 report

Code
github.com/connectwithvanshika/TensorTitans_SectionC